SAAD DAHLAB UNIVERSITY BLIDA
FACULTY OF TECHNOLOGY
COMMON CORE DEPARTEMENT: ST

P.W N°1
CALCULATION OF UNCERTAINTIES

I. THEORETICAL STUDY

A physical measurement never gives the exact value of the quantity being measured. Its result is only approximated by a number of uncertainties.
There are generally three causes of uncertainty:

- The uncertainties of the measuring device.
- Fault in the measurement method.
- Human limitations (problems reading analogue device).

I.1 Uncertainty assessment:

Two types of quantities are used to assess uncertainties qualitatively and quantitatively:

a. Absolute uncertainty:

It represents the absolute value of the difference between the measured value $X_{measured}$ and the exact value of the quantity X_{exact}. It has the same unit as the quantity. It is noted ΔX.

$$\Delta X = \left| X_{exact} - X_{mesured} \right|$$

As ΔX depends on the exact value, which we do not know, we can only estimate its upper limit.

***Example**: The length of an object is 153mm to within 2mm: this means that the result of the measurement is 153mm, but the study of the causes of uncertainty leads us to believe that the exact value cannot deviate from this value by more than 2mm:*

$$151mm \leq \ exact\ length\ (L) \leq 155mm$$
$$L = (153 \pm 2)\,mm$$
$$L \in [153-2, 153+2]\,mm$$

b. Relative uncertainty:

Clearly, an uncertainty of 1mm does not have the same significance when it relates to a length of 10m or a length of 1cm. This has led us to define the relative uncertainty as the ratio of the absolute uncertainty to the value taken as the result of the measured quantity, i.e.:

$$\text{Relative uncertainty} = \frac{\Delta X}{X_{mesured}}$$

This uncertainty has no unit and is expressed in %.

I.2 Different sources for evaluating absolute uncertainty:

a. ***Systematic error*** ΔX_{syst}: These errors are due to imperfections in the graduations or settings (inaccurate calibration of the weights on a balance, of the scale on an ammeter). These errors affect the result constantly and in the same direction.

b. ***Random errors or reading errors*** ΔX_{read}: these are linked to the observer and are sometimes difficult to assess. We try to reduce them by taking several measurements and averaging the results observed.

c. ***Instrumentation errors*** ΔX_{instr}: these are given by the manufacturer as follows:

$$\Delta X_{instr} = \frac{Classe \times Caliber}{100}$$

I.3 Calculating uncertainties:

There are two ways of measuring a quantity: either it is measured directly using a device (a graduated ruler, a stopwatch, i.e.....), or it is calculated according to a relationship that includes the quantity measured.

I.3.1 Direct measurement:

Measurement taken by a calibrated instrument.

a. ***Case of several measurements:***

Random errors occur in both directions. To eliminate them, ***N*** measurements of the same physical quantity X are taken. The result that is closest to the exact value is given by the mean value:

$$X_{avg} = \frac{X_1 + X_2 + X_3 + ... + X_n}{N} = \frac{\sum_{i=1}^{N} X_i}{N}$$

To assess uncertainties, the following differences are formed:

$\varepsilon i = X_i - X_{avg}$ with: X_i: partial measurement
X_{avg}: average measurement

The sign of εi can be positive or negative, and we take its absolute value: $\varepsilon i = |X_i - X_{avg}|$
and

$$\Delta X = \text{upper } \varepsilon_i = \text{upper } |X_i - X_{avg}|$$

So:

$$X = X_{avg} \pm \Delta X$$

Example:

During a practical session, three students measure the edge l of a cube:

Student 1	Student 2	Student 3
5.30 cm	5.60 cm	5.40 cm

Calculating the average value of the edge *l* gives them: 5.46 cm and the absolute uncertainty is:

$$\Delta l = upper\left\{\left|0,16\right|,\left|-0,14\right|,\left|0,06\right|\right\} = 0,16cm$$

After rounding, they will present their results as follows: $l = (5,5 \pm 0,2)cm$

The exact edge of the cube belongs to the interval: $l \in [5,3;5,7]cm$

and the relative error is: $\Delta l / l = 4\%$

***b. Case of a single measure*:**

The measurement is carried out once and the absolute uncertainty is taken to be

$$\Delta X = \Delta X_{syst} + \Delta X_{inst} + \Delta X_{read}$$

The relative uncertainty will be:

$$\frac{\Delta X}{X_{mesured}}$$

I.3.2 Indirect measurement:

A measurement is said to be indirect if it is deduced using a mathematical relation. For example, the velocity of an object is an indirect measurement because it is determined by measuring the distance and the corresponding time. There are two methods for determining the uncertainty of an indirect measurement: the differential method and the logarithmic method.

a) The differential method

Given a quantity $X = f(x,y,z)$ determined by the measurement of x,y and z. The latter are affected by uncertainties Δx, Δy and Δz, we want to know ΔX the uncertainty involved in determining f, that is:

$$\Delta X = \Delta f = \text{absolute uncertainty} = f_{measured} - f_{exact}$$

Since the uncertainties are small, we approximate Δf by df the differential of f and dx, dy and dz by Δx, Δy and Δz.

The differential of f is:

$$df = \left(\frac{\partial f}{\partial x}\right)dx + \left(\frac{\partial f}{\partial y}\right)dy + \left(\frac{\partial f}{\partial z}\right)dz$$

We'll have:

$$\Delta f = \left|\frac{\partial f}{\partial x}\right|\Delta x + \left|\frac{\partial f}{\partial y}\right|\Delta y + \left|\frac{\partial f}{\partial z}\right|\Delta z$$

The absolute value of Δf is increased by the sum of the absolute values, thereby overestimating the uncertainty of Δf.

$$\Delta f \le \left|\frac{\partial f}{\partial x}\right|\Delta x + \left|\frac{\partial f}{\partial y}\right|\Delta y + \left|\frac{\partial f}{\partial z}\right|\Delta z$$

Example :

Let $f(x,y,z) = xy\,z$

$$\frac{\partial f}{\partial x} = yz\ ,\frac{\partial f}{\partial y} = xz\ \ ,\frac{\partial f}{\partial z} = xy$$

$$df = (yz)dx + (xz)dy + (xy)dz$$

$$\frac{df}{f} = \left(\frac{yz}{xyz}\right)dx + \left(\frac{xz}{xyz}\right)dy + \left(\frac{xy}{xyz}\right)dz$$

$$\frac{df}{f} = \frac{dx}{x} + \frac{dy}{y} + \frac{dz}{z}$$

$$\frac{\Delta f}{f} = \frac{\Delta x}{x} + \frac{\Delta y}{y} + \frac{\Delta z}{z}$$

b) The logarithmic method

In practice, it is often preferable to use the relative uncertainty, which can be expressed as a percentage, and there is a practical method for estimating this without having to calculate Δf:

- Calculate the natural logarithm of f;
- Calculate the logarithmic differential;
- We move on to the absolute values of the relative uncertainties.

Example:

$f(x,y,z) = \dfrac{xy^2}{z}$; x,y and z are measured quantities and Δx, Δy and Δz their respective absolute uncertainties. Let's calculate the relative uncertainty of f:

$$ln\,f = ln\left(\frac{xy^2}{z}\right) = ln\,x + ln\,y^2 - ln\,z = ln\,x + 2\,ln\,y - ln\,z$$

$$d\left(ln\,f\right) = \frac{df}{f} = \frac{dx}{x} + 2\frac{dy}{y} - \frac{dz}{z}$$

$$\frac{\Delta f}{f} = \frac{\Delta x}{x} + 2\frac{\Delta y}{y} + \frac{\Delta z}{z}$$

I.4 Uncertainty and significant numbers

An experimental measurement is always affected by an uncertainty, whatever the quality of the equipment or the skill of the experimenter. If the measurement of a length is accurate to within 0.2 cm, there is no point in writing L = 11.7438± 0.2 cm. It is sufficient to write L = (11.7± 0.2) cm.

Frequent error: we measure a diameter d = (10.0± 0.1) cm of a circle and deduce the perimeter P.

False answer: $P = \pi\, d$ = 31.4159265± 0.314159 cm.

Correct answer: P = (31.4 ± 0.3) cm

First and last name:	**Group:** .....

II. WORK REQUIRED:

To calculate the perimeter (P) and area (A) of a rectangle, we repeated the measurements of its length (L) and width (w) three times. We obtained the following values in cm:

L (cm)			
w (cm)			

1. Determine the average value of L:

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$$L_{avg} = \qquad cm$$

2. Determine the average value of w:

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$$w_{avg} = \qquad cm$$

3. Determine the absolute uncertainty on the measurement of L:

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$$\Delta L = \qquad cm$$

4. Determine the absolute uncertainty on the measurement of w:

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$$\Delta w = \qquad cm$$

5. Give the value of the perimeter P.

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$$P = \qquad cm$$

6. Give the value of the area A.

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$$A = \qquad\qquad cm^2$$

7. Determine the absolute uncertainty of P (ΔP) by using the differential method

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$$\Delta P = \qquad\qquad cm$$

8. Deduce the relative uncertainty: $\frac{\Delta P}{P}$

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$$\frac{\Delta P}{P} = \qquad = \qquad \%$$

9. Determine the absolute uncertainty of P (ΔP) by using the logarithmic method

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$$\frac{\Delta P}{P} = \qquad = \qquad \%$$

Then:

$$\Delta P = \qquad\qquad cm$$

10. Give the exact value of P as:

$$P = (\qquad \pm \qquad)cm$$

11. Determine the absolute uncertainty of A (ΔA) by using the differential method

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$$\Delta A = \qquad\qquad cm^2$$

12. Deduce the relative uncertainty: $\frac{\Delta A}{A}$

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$$\frac{\Delta A}{A} = \qquad = \qquad \%$$

13. Determine the absolute uncertainty of A (ΔA) by using the logarithmic method

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$$\frac{\Delta A}{A} = \qquad = \qquad \%$$

Then:

$$\Delta A = \qquad\qquad cm^2$$

14. Give the exact value of A as:

$$A = (\qquad \pm \qquad)cm^2$$